B5.1 Read transaction flows

This section gives examples of the interconnect protocol flow for Read transactions.

B5.1.1 Read transactions with DMT and without snoops

For Read transactions without snoops, it is recommended to use DMT.

Figure B5.1 shows an example DMT transaction flow using the ReadShared transaction.

In Figure B5.1, a response from SN-F to HN-F is not required because CompAck from the Requester is used to deallocate the request at Home.

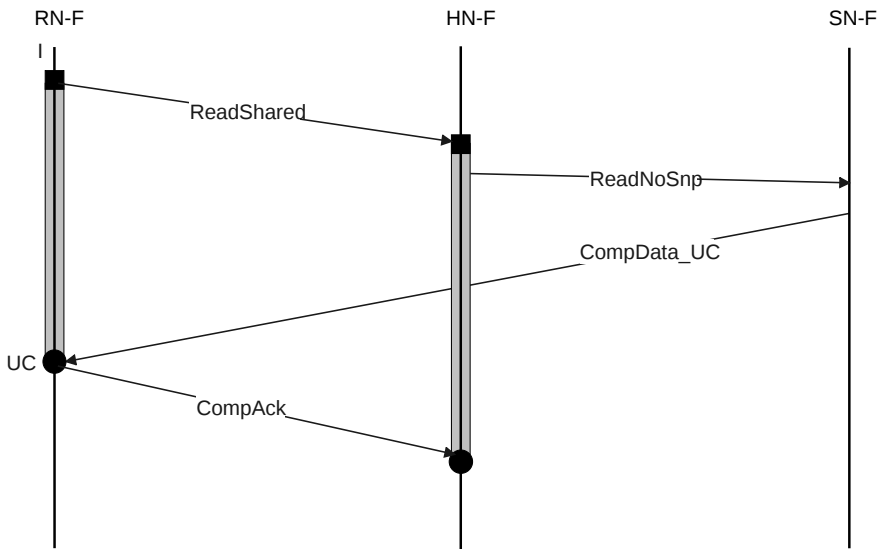

Figure B5.1: DMT Read transaction example without snoops

The steps in the ReadShared transaction flow in Figure B5.1 are:

1. RN-F sends a ReadShared request to HN-F.
2. HN-F sends a ReadNoSnp request to SN-F.
3. SN-F sends a data response directly to RN-F, using CompData_UC.
4. RN-F sends CompAck to HN-F as the request is ReadShared and requires CompAck to complete the transaction.

B5.1.2 Read transaction with DMT and with snoops

For Read transactions with snoops and data from memory, DMT is recommended.

Figure B5.2 shows an example DMT transaction flow using the ReadShared transaction.

From SN-F to HN-F, a response is not required because CompAck from the Requester is used to deallocate the request at Home.

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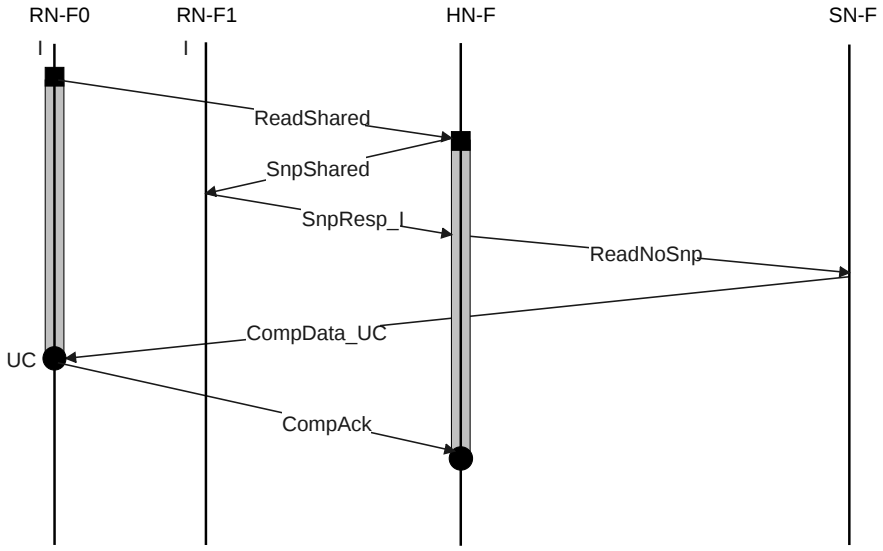

Figure B5.2: DMT Read transaction example with snoops and data from memory

The steps in the ReadShared transaction flow in Figure B5.2 are:

1. RN-F0 sends a ReadShared request to HN-F.
2. HN-F sends a SnpShared request to RN-F1. RN-F1 returns SnpResp_I response to HN-F.
3. HN-F sends a ReadNoSnp request to SN-F after receiving the Snoop response from RN-F1. This guarantees that RN-F1 has not responded with data.
4. SN-F sends a data response directly to RN-F0, using CompData_UC.
5. RN-F0 sends CompAck to HN-F as the request is ReadShared and requires CompAck to complete the transaction.

B5.1.3 Read transaction with DCT

For Read transactions with snoops and data from other Request Node caches, Direct Cache Transfer (DCT) is recommended.

B5.1.3.1 DCT from cache line in UC state

Figure B5.3 shows an example flow for a DCT transaction. The Requester is RN-F0 and the forwarding cache is at RN-F1.

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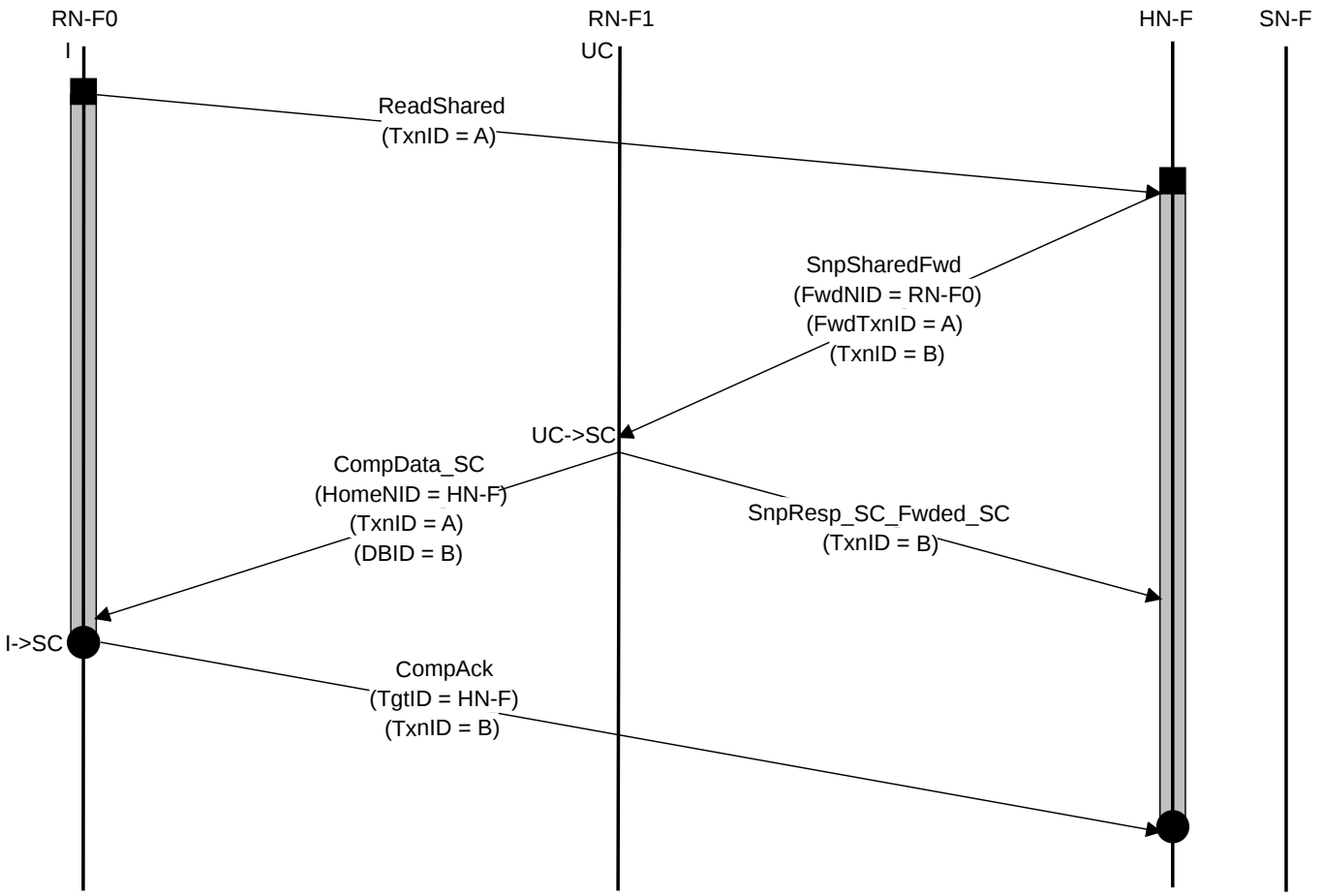

Figure B5.3: DCT from cache line in UC state

The steps in the DCT transaction flow in Figure B5.3 are:

1. RN-F0 sends a ReadShared request to HN-F.
2. HN-F sends a SnpSharedFwd, a Forwarding snoop request to RN-F1.
3. RN-F1 forwards CompData_SC response to RN-F0. The TxnID is the same as the original ReadShared request.
4. RN-F1 also sends a SnpResp_SC_Fwded_SC snoop response to HN-F. RN-F1 cache line state transitions from UC to SC.
5. After receiving the CompData_SC response, RN-F0 cache line state transitions from I to SC and RN-F0 sends a CompAck response to HN-F.

Note

Steps 3 and 4 in the DCT transaction flow can occur in any order as CompData and SnpResp are sent on different channels.

B5.1.3.2 Double data return in a DCT transaction

Figure B5.4 shows an example DCT transaction flow that sends the data to HN-F and forwards the data to the RN-F0.

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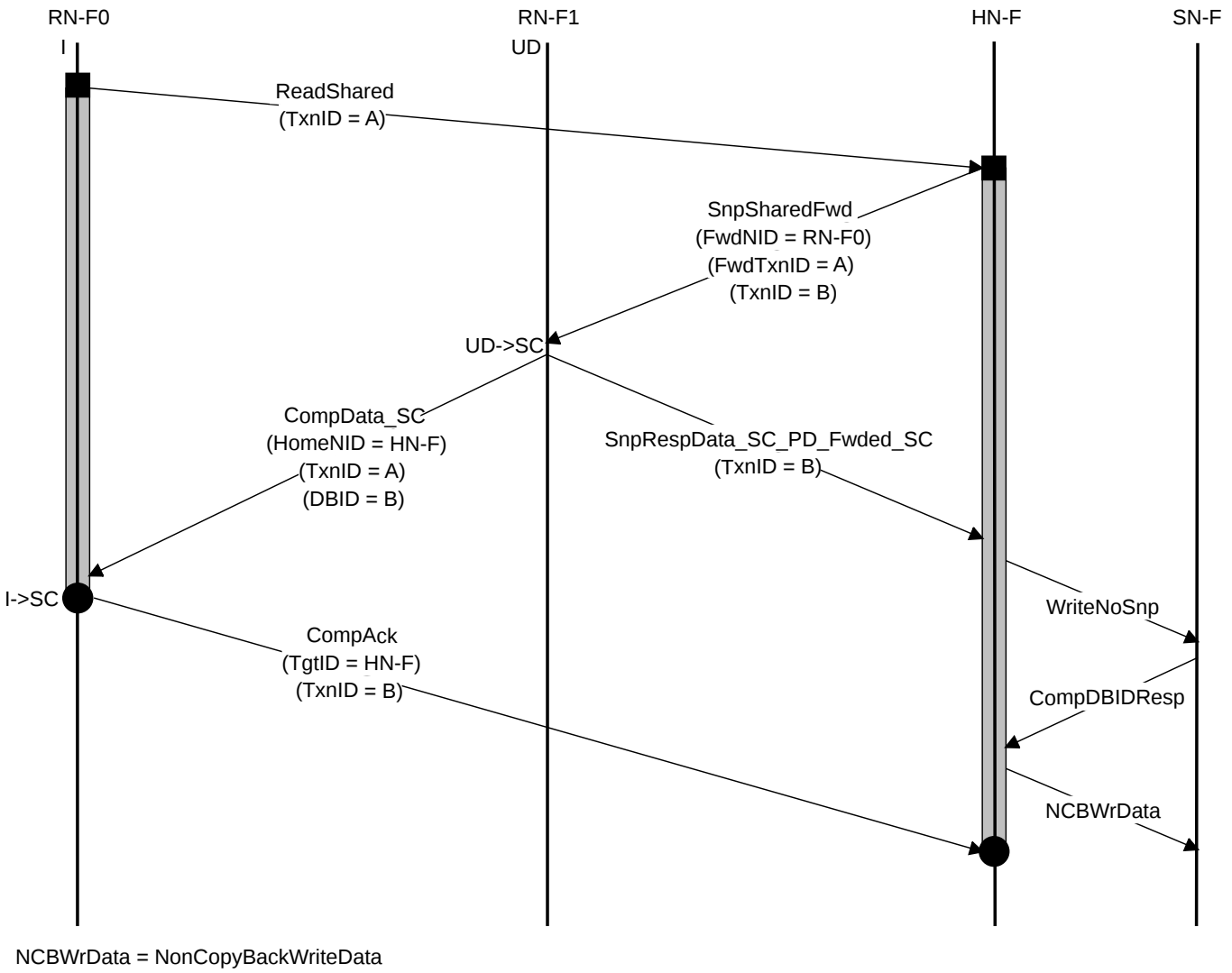

Figure B5.4: Double data return in DCT transaction

The steps in the DCT transaction flow in Figure B5.4 are:

1. RN-F0 sends a ReadShared request to HN-F.
2. HN-F sends a SnpSharedFwd Snoop request to RN-F1.
3. RN-F1 sends CompData_SC response to RN-F0. The TxnID is the same as the original ReadShared request. RN-F0 cache line transitions from I to SC.
4. RN-F1 also sends a SnpRespData_SC_PD_Fwded_SC snoop response to HN-F that includes a copy of the cache line and passes responsibility for the Dirty cache line to HN-F. RN-F1 cache line transitions from UD to SC.
5. HN-F sends a WriteNoSnp request to the SN-F.
6. SN-F responds to the HN-F with CompDBIDResp to request the data.
7. HN-F sends NCBWrData to the SN-F.
8. RN-F0 sends CompAck after receiving the Data response.

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B5.1.4 Read transaction without DMT or DCT

Figure B5.5 shows an example of the flow without DMT using the ReadNoSnp transaction. In Figure B5.5, the ReadNoSnp has the ExpCompAck set in the original request.

The request does not generate any snoops and receives the data from a response to a memory read by the HN-F.

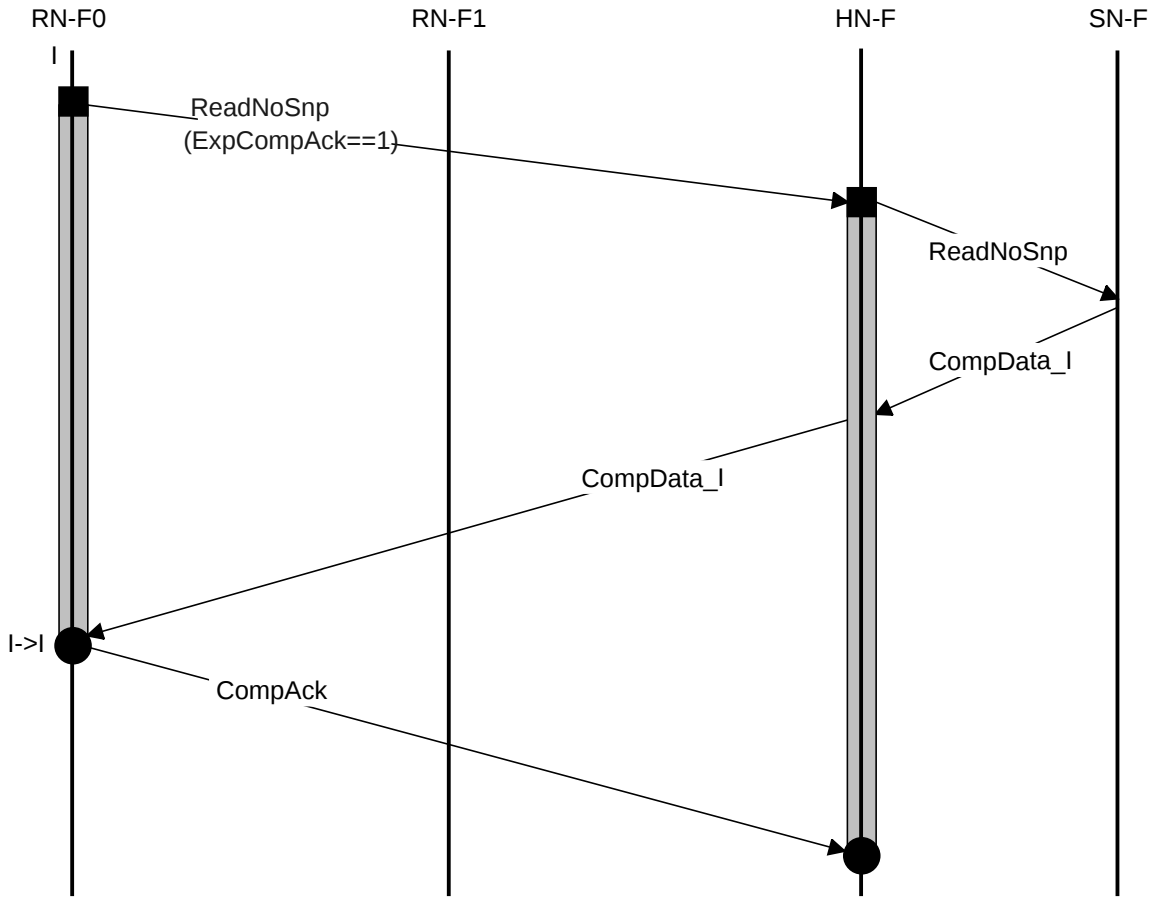

Figure B5.5: ReadNoSnp transaction flow

The steps in the ReadNoSnp transaction flow in Figure B5.5 are:

1. RN-F0 issues a ReadNoSnp transaction, with ExpCompAck set to 1.
2. HN-F receives and allocates the request.

> **Note**
>
> HN-F does not send snoops as the request is recognized as a Non-snoopable request type.

3. HN-F sends a ReadNoSnp to SN-F.
4. SN-F returns CompData_I to HN-F.
5. HN-F in turn returns the data to RN-F0.

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> **Note**
>
> If ExpCompAck is 0 in the original ReadNoSnp request, the HN-F could have deallocated the request at this point.

6. RN-F0 deallocates the request and sends CompAck towards the HN-F.
7. HN-F receives the CompAck response and deallocates the request.

B5.1.5 Read transaction with snoop response with partial data and no memory update

An example of this type of flow is a ReadUnique transaction.

Figure B5.6 shows the transaction flow.

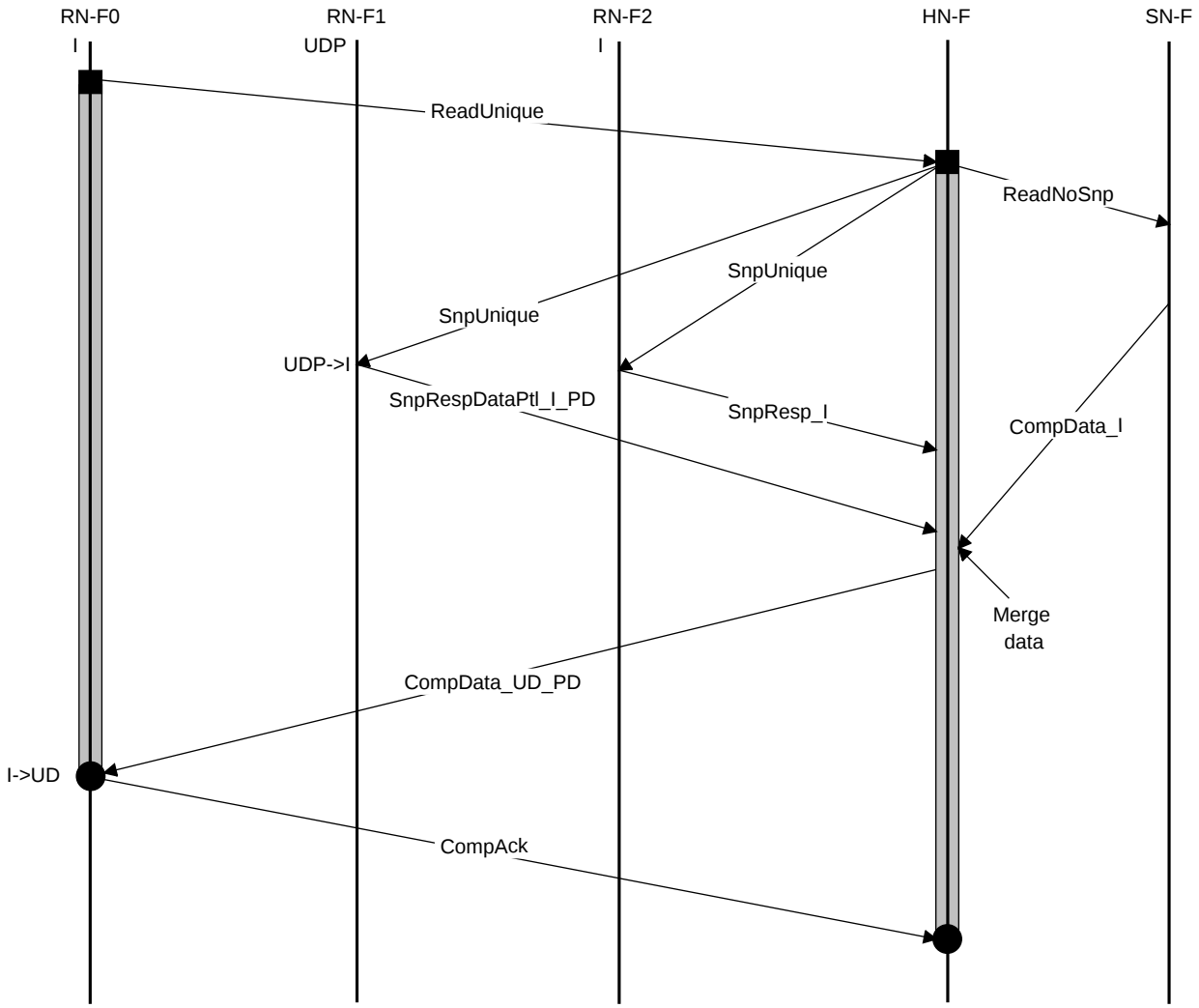

Figure B5.6: ReadUnique with partial data snoop response

The steps in the ReadUnique with partial data snoop response transaction flow in Figure B5.6 are:

1. RN-F0 sends ReadUnique request to HN-F.
2. HN-F sends ReadNoSnp request to SN-F and SnpUnique requests to RN-F1 and RN-F2.

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3. RN-F1 transitions the cache line from UDP to I and returns SnpRespDataPtl_I_PD to HN-F. RN-F2 returns SnpResp_I to the HN-F. Meanwhile, the SN-F returns CompData response to HN-F. HN-F merges the data returned from the SN-F with the partial data received from the RN-F1.
4. HN-F sends CompData_UD_PD to RN-F0. RN-F0 cache line state transitions from I to UD.
5. RN-F0 issues CompAck response to the HN-F to indicate transaction completion.

B5.1.6 Read transaction with snoop response with partial data and memory update

An example of this type of flow is a ReadClean transaction.

Figure B5.7 shows the transaction flow.

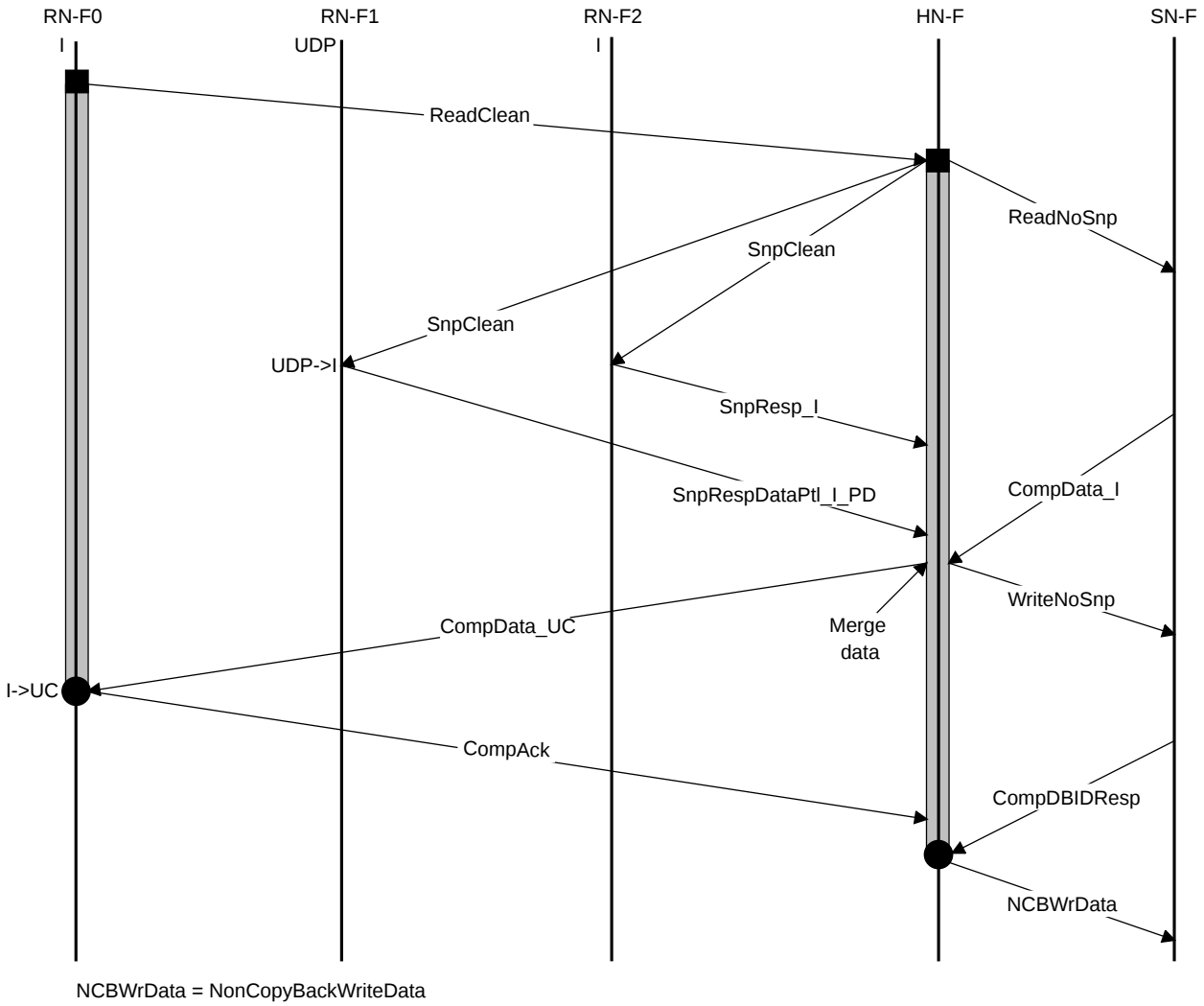

Figure B5.7: ReadClean with partial data snoop response

The steps in the ReadClean with partial data snoop response transaction flow in Figure B5.7 are:

1. RN-F0 sends ReadClean request to HN-F.
2. HN-F sends ReadNoSnp request to SN-F and SnpClean requests to RN-F1 and RN-F2. RN-F1 cache line state transitions from UDP to I.

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3. RN-F1 returns SnpRespDataPtl_I_PD and RN-F2 returns SnpResp_I to the HN-F. Meanwhile, the SN-F returns CompData_I response to HN-F. This merges the data.
4. HN-F sends CompData_UC to RN-F0 and WriteNoSnp to SN-F. RN-F0 cache line state transitions from I to UC.
5. RN-F0 issues CompAck response to the HN-F to indicate transaction completion.
6. SN-F issues CompDBIDResp to HN-F.
7. HN-F sends NCBWrData to SN-F.

B5.1.7 ReadOnce* and ReadNoSnp with early Home deallocation

Figure B5.8 shows the optimized flow for an unordered ReadOnce request.

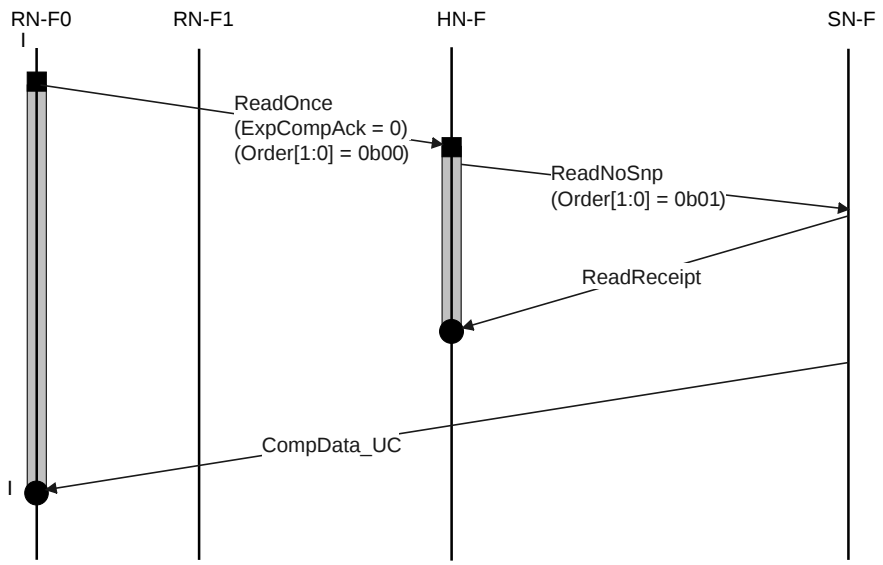

Figure B5.8: DMT optimization for an unordered ReadOnce

The steps in the optimized ReadOnce transaction flow in Figure B5.8 are:

1. RN-F0 sends an unordered ReadOnce request to HN-F with Order[1:0] set to `0b00`.
2. HN-F sends a DMT ReadNoSnp request to SN-F with the Order[1:0] set to `0b01`.
3. SN-F sends ReadReceipt to Home.
4. HN-F deallocates the request after receiving the ReadReceipt response.
5. SN-F sends CompData_UC directly to RN-F0.

Note

Use of a ReadNoSnp transaction from Home to Subordinate, in the case where CompAck is not required, avoids the need to send a RespSepData response from Home to Requester.

B5.1.8 ReadNoSnp transaction with DMT and separate Non-data and Data-only response

Figure B5.9 shows an example DMT transaction flow with separate Non-data and Data-only response.

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In this example, there is no ordering requirement and HN-F can deallocate the request once it receives ReadReceipt, without waiting for CompAck from the RN-F.

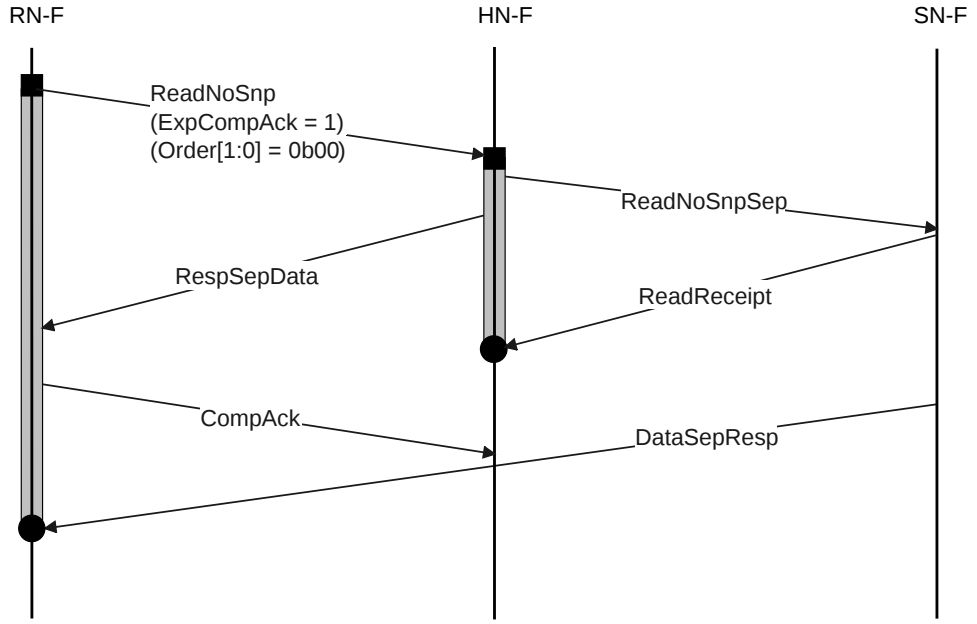

Figure B5.9: DMT Read transaction example with separate Non-data and Data-only

The steps in the ReadNoSnp transaction in Figure B5.9 are:

1. RN-F sends an unordered ReadNoSnp request to HN-F.
2. HN-F sends a ReadNoSnpSep request to SN-F.
3. HN-F sends a RespSepData response to RN-F.
4. SN-F sends a ReadReceipt to HN-F.
5. RN-F sends CompAck after receiving RespSepData.
6. SN-F sends DataSepResp to RN-F returning the read data.

B5.1.9 ReadNoSnp transaction with DMT with ordering and separate Non-data and Data-only

Figure B5.10 shows an example DMT transaction flow with ordering and separate Non-data and Data-only.

ReadNoSnp with non-zero Order field in Figure B5.10 requires that:

- Next ordered request can be sent only after receiving of RespSepData.
- RN-F must wait for RespSepData and at least one packet of DataSepResp before sending CompAck.
- HN-F must not send next ordered request to SN-F until receiving CompAck.

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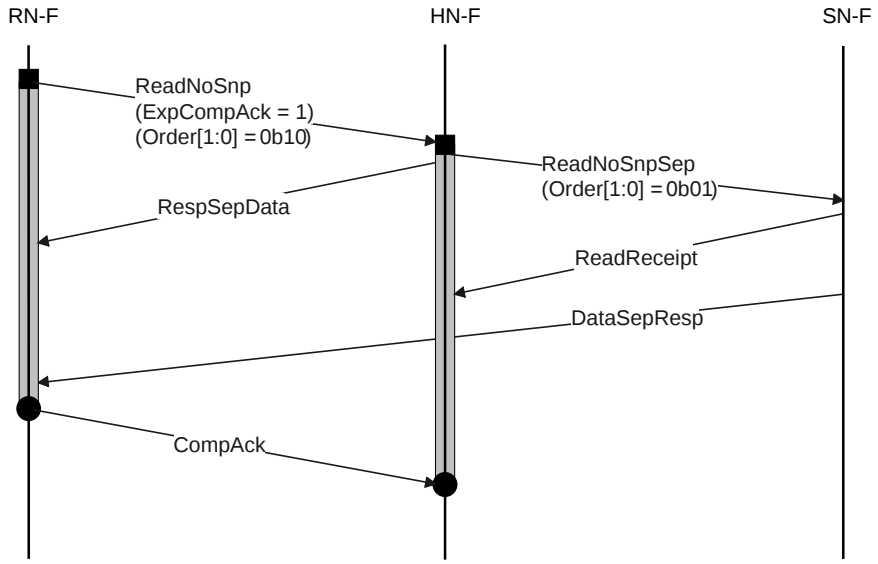

Figure B5.10: DMT Read transaction example with ordering and separate Non-data and Data-only

The steps in the Read transaction with ordering and separate Non-data and Data-only in Figure B5.10 are as follows:

1. RN-F sends ReadNoSnp to HN-F, with ExpCompAck set to 1. Order[1:0] is set to `0b10`.
2. HN-F sends ReadNoSnpSep to SN-F with Order[1:0] set to `0b01`.
3. HN-F returns RespSepData to RN-F.
4. SN-F returns ReadReceipt to HN-F.
5. SN-F sends DataSepResp to RN-F directly.
6. RN-F issues CompAck to HN-F.

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